AWS Lab: IAM Least Privilege Re-scoping for Operational Service Role

Duration : 30 Min.

Scenario

A nighttime operations service role 'svc-nightops' currently has 'AdministratorAccess' attached. An internal audit flags this as a critical security finding. The role only requires permissions to describe EC2 instances, restart a specific tagged instance (Environment=NightlyTest), and stream CloudWatch logs. The engineer must author a scoped IAM policy adhering to the principle of least privilege.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** AWS Identity and Access Management (IAM) - **Target Service Role:** `svc-nightops` - **Required Permissions:** `ec2:DescribeInstances`, `ec2:RebootInstances` (scoped ARN), `logs:CreateLogStream` - **Security Constraint:** Elimination of `Action: '*'` and `Resource: '*'`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Inspect current permissions for svc-nightops in /opt/aws_iam/svc_nightops_policy.json. 2. Apply and attach least-privilege IAM policy directly in AWS IAM granting ec2:DescribeInstances, ec2:RebootInstances on specific ARN, and logs:CreateLogStream. 3. Ensure Action: *'' and Resource: ''* are completely eliminated.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Audit identification: recognize over-permissioned AdministratorAccess grant	4 Marks
TC2	Policy authoring: write valid JSON IAM policy with scoped Actions and Resource ARNs	3 Marks
TC3	Least privilege compliance: verify zero administrative wildcard permissions exist	3 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.